Link Layer & LANs Formula Sheet (Chapter 6)

1. Error Detection & Correction

Cyclic Redundancy Check (CRC)

An algorithm for calculating ‘r‘ error-detection bits, ‘R‘, for a block of ‘d‘ data bits, ‘D‘.
1. Sender and receiver must agree on an ‘r+1‘-bit generator pattern, ‘G‘.
2. The sender calculates the remainder of the following binary division (using XOR for subtraction):

$$R = \text{remainder}\left(\frac{D \cdot 2^r}{G}\right)$$

3. The sender transmits the data ‘D‘ followed by the remainder ‘R‘.
4. The receiver divides the entire received ‘d+r‘ bit frame by ‘G‘. If the remainder is non-zero, an error is detected.

2. Multiple Access Protocol Efficiency

Efficiency is defined as the long-run fraction of time during which frames are being transmitted on the channel successfully (without collision).

Slotted ALOHA Efficiency

Calculates the probability of a successful transmission in a single time slot in a network with N nodes, where each node transmits with probability ‘p‘.

$$P(\text{success by a specific node}) = p \cdot (1-p)^{N-1}$$

This is the probability that this specific node transmits (‘p‘) AND all ‘N-1‘ other nodes do NOT transmit (‘(1-p)$^{N-1}$‘).

$$\text{Efficiency} = N \cdot p \cdot (1-p)^{N-1}$$

This is the probability that any one of the N nodes successfully transmits.
Maximum Efficiency: As N approaches infinity, the maximum possible efficiency of Slotted ALOHA is $1/e \approx \mathbf{0.37}$.

Pure (Unslotted) ALOHA Maximum Efficiency

The maximum efficiency is half that of Slotted ALOHA due to a larger ”vulnerable period” for collisions.

$$\text{Max Efficiency} = \frac{1}{2e} \approx \mathbf{0.18}$$

3. CSMA/CD (Ethernet) Performance

CSMA/CD Efficiency

The efficiency of classic Ethernet with CSMA/CD depends on the ratio of the maximum propagation delay to the frame transmission time.

$$\text{Efficiency} = \frac{1}{1 + 5 \cdot \frac{d_{prop}}{d_{trans}}}$$

d_{prop}**:** The maximum propagation delay between any two nodes on the shared bus.
d_{trans}**:** The time to transmit a maximum-sized frame.
How to Use: This formula shows that as the transmission time gets much larger than the propagation time (i.e., for faster links or smaller networks), the efficiency approaches 100%. Conversely, as propagation delay becomes dominant (e.g., in a very large, slow network), efficiency drops.

4. Switch Table Capacity

The number of entries required in a switch's self-learning table depends on the number of hosts/nodes connected to the LAN.

Switch Table Size

If a LAN has 'K' hosts and routers connected to it, the switch tables in that LAN will need to be able to store up to 'K' entries.
TTL (Time To Live): While switch tables can theoretically grow to the size of the LAN, entries are typically aged out after a period of inactivity (e.g., 20 minutes). This means the actual number of entries at any given time is the number of hosts that have been active recently.